Research Article Vol. 1, No. 12 / 15 Dec 2022 / Optics Continuum 2600
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Zernike mode rescaling extends capabilities of adaptive optics for microscopy

JAKUB CZUCHNOWSKI[1,2,*] AND ROBERT PREVEDEL[1,3]

[1] *Cell Biology and Biophysics Unit, European Molecular Biology Laboratory, Heidelberg, Germany*
[2] *Collaboration for joint PhD degree between EMBL and Heidelberg University, Faculty of Biosciences, Germany*
[3] *robert.prevedel@embl.de*
[*] *jakub.czuchnowski@embl.de*

Abstract: Zernike polynomials are widely used mathematical models of experimentally observed optical aberrations, and they have found widespread use in adaptive optic realizations that are used to correct wavefront aberrations. However, Zernike aberrations lose their orthogonality when used in combination with Gaussian beams and, as a consequence, start to cross-couple between each other, a phenomenon that does not occur for Zernike aberrations in plane waves. Here, we describe how the aberration radius (i.e. the radius of the beam relative to the active aperture of an active optical element) influences this cross-coupling of Zernike aberrations in a way that is distinct from simple truncation or balancing. Furthermore, we show that this effect can actually be harnessed to allow efficient compensation of higher-order aberrations using only low-order Zernike modes. This finding has important practical implications, as it suggests the possibility of using adaptive optics devices with low element numbers to compensate aberrations that would normally require more complex and expensive devices.

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1. Introduction

Adaptive optics (AO) allowed breakthrough discoveries in astronomy by enabling telescopic observation through difficult atmospheric conditions [1]. These breakthroughs were enabled by important developments in understanding and tackling optical aberrations introduced into light during its propagation. In recent years, developments of AO for microscopic imaging allowed for high resolution visualisation of structures deep in scattering materials (e.g. biological tissues) [2,3]. These, however, require dedicated hardware different from the one used in astronomy, but maybe even more importantly, also different theoretical frameworks suited to the conditions used in microscopy (e.g. use of Gaussian beams for laser microscopy). Currently, the dominant theoretical model of optical aberrations in microscopy are Zernike polynomials due to their orthogonal nature and isomorfisms to experimentally observable aberrations [4].

However, previous work has shown that Zernike aberrations are not orthogonal in the case of Gaussian beams and can display significant cross-coupling as described both by the Strehl ratio approach [5,6] as well as by evaluating coupling into higher-order Laguerre-Gauss (LG) modes [7]. While more optimised basis sets for experimental AO have indeed been proposed [8], here we pursue an alternative approach that actually harnesses the non-ortogonality of Zernike aberrations.

In particular, by analysing the interactions of weak aberrations we show that by manipulating the Zernike mode size with relation to the Gaussian beam diameter (e.g. by changing the ratio between the active aperture of an AO element and the size of the beam) it is possible to strongly enhance cross-coupling properties of Zernike aberrations which could be used in practice to increase the correction capabilities of commonly used AO elements. This represents a novel approach that goes beyond previously analysed effects of truncation [5] and aberration balancing

#475170 https://doi.org/10.1364/OPTCON.475170
Journal © 2022 Received 6 Sep 2022; accepted 28 Oct 2022; published 14 Dec 2022

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using Zernike-Gauss polynomials [9] by allowing superpositions of rescaled Zernike aberrations which can be experimentally realised by a combination of multiple deformable mirrors.

2. Effects of aberration radius on coupling between Zernike aberrations and Laguerre-Gauss modes

Laguerre-Gauss beams are inherently orthogonal and therefore do not cross-couple between each other in free space propagation. However, Zernike type aberrations are capable of inducing energy coupling between different LG-beams [10]. The coupling coefficient between LG-modes induced by a particular Z_n^m aberration can be described as:

$$k_{p,l,p',l'}^{n,m} = \int_A LG_{p,l} \exp(ikZ_n^m) LG_{p',l'}^* dA \tag{1}$$

where:

$$Z_n^m(r,\phi) = \begin{cases} A_n^m R_n^m(r)\cos(m\phi) & \text{for } m \geqslant 0 \\ A_n^m R_n^{|m|}(r)\sin(|m|\phi) & \text{for } m<0, \end{cases} \tag{2}$$

$$LG_{lp}(r,\phi,z,k) = C_{lp}^{LG}\left(\frac{r\sqrt{2}}{w(z)}\right)^{|l|} L_p^{|l|}\left(\frac{2r^2}{w^2(z)}\right)\exp(-il\phi)G(r,z,k), \tag{3}$$

where k denotes the wavenumber and A is the pupil area over which the beam is integrated (see Supplement 1 Section S1 for full definitions). In the weak aberration regime the phase aberration can be expressed as:

$$\exp(ikZ_n^m) \approx 1 + ikZ_n^m. \tag{4}$$

This enables Eq. (1) to be solved analytically [10]:

$$k_{p,l,p',l'}^{n,m} = \int_0^{2\pi}\int_0^R LG_{p,l} LG_{p',l'}^* (ikZ_n^m) r dr d\phi = \delta_{p,p'}\delta_{l,l'} + I_\phi I_r \tag{5}$$

where I_ϕ and I_r are the azimuthal and radial integral respectively. The azimuthal part (I_ϕ) determines the coupling condition (see [7,10,11] and Supplement 1 Section S2 for details) and provides a mapping between particular groups of Zernike aberrations and LG-modes. However, the radial integral (I_r) determines coupling within the groups of LG-modes determined by Equation S9 effectively regulating the degree of cross-compensation [10,11]:

$$\begin{aligned}[I_r]_{p,l,p',l'}^{n,m} =& A_n^m \frac{ik}{\pi}\sqrt{p!p'!(p+|l|)!(p'+|l'|)!}\exp(i\Delta o\psi) \\ &\times \sum_{i=0}^{p}\sum_{j=0}^{p'}\sum_{h=0}^{\frac{1}{2}(n-m)} \frac{(-1)^{i+j+h}}{(p-i)!(p'-j)!(|l|+i)!(|l'|+j)!i!j!}\frac{1}{\hat{R}^{\frac{1}{2}(n-2h)}}\times \\ &\frac{(n-h)!}{(\frac{n+m}{2}-h)!(\frac{n-m}{2}-h)!h!}\gamma(i+j-h+\frac{1}{2}(|l|+|l'|+n)+1,\hat{R})\end{aligned} \tag{6}$$

where A_n^m is the amplitude of the Zernike aberration, ψ is the Gouy phase described by Equation S7, $\hat{R} = \frac{2R^2}{w^2}$ encodes the ratio between the Zernike radius (R) and the beam diameter (w), $\Delta o = 2p + l - 2p' - l'$ is the difference in orders between incident and coupled LG-modes and $\gamma(a,x) = \int_0^x t^{a-1}e^{-t}dt$ is the lower incomplete gamma function. As can be appreciated from Eq. (6) the relation between the coupling coefficient and the ratio between the Zernike radius and the beam diameter ($\hat{R}$) is a complex one and does not allow for straightforward analytical analysis. Therefore we have used this framework to numerically evaluate the effects of aberration radius on the beam properties. One can appreciate that the coupling distribution into higher order

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LG-modes strongly depends on the ratio between the aberration and beam radii (R/w) for both radially symmetric (Fig. 1(B)) and asymmetric Zernike modes (Supplementary Figure S1A). To check whether this has an effect on Zernike cross-compensation [7] we calculate the recoverable power fraction (P_{rec}) for beams aberrated with different R/w ratios.

$$P_{rec} = \frac{\sum_i \left(k_i^{n_p,m_p}\left(\frac{R_p}{w}\right) + \alpha k_i^{n_a,m_a}\left(\frac{R_a}{w}\right)\right)\overline{\left(k_i^{n_p,m_p}\left(\frac{R_p}{w}\right) + \alpha k_i^{n_a,m_a}\left(\frac{R_a}{w}\right)\right)}}{1 - \left|k_{0,0,0,0}^{n_p,m_p}\left(\frac{R_p}{w}\right)\right|^2} \tag{7}$$

where α denotes the optimal correction amplitude which can be calculated from Equation S11, $i = (p, l, p', l')$ excluding $(p', l') = (0, 0)$. By setting $(n_a, m_a) = (n_p, m_p)$ we explore the self-compensation of Zernike aberration with different R/w ratios. We show that the recoverable power decreases (Fig. 1(C), Supplementary Figure S1B) which implies that rescaled Zernike modes (with $R/w \neq 1$) are able to compensate the native Zernike aberrations (with $R/w = 1$) only to a limited degree (which is intuitive since rescaling reduces the mode self-similarity which can be observed in Fig. 1(A)). On the other hand, the recoverable power between different aberrations ($(n_a, m_a) \neq (n_p, m_p)$) increases which facilitates stronger cross compensation between the modes (Fig. 1(D), Supplementary Figure S1C). Interestingly, reducing the R/w ratio allows Zernike modes to more efficiently cross-compensate higher-order aberrations which has important practical implications as it suggests it's possible to extend aberration correction capabilities of AO elements (e.g. deformable mirrors) beyond their specified limitations.

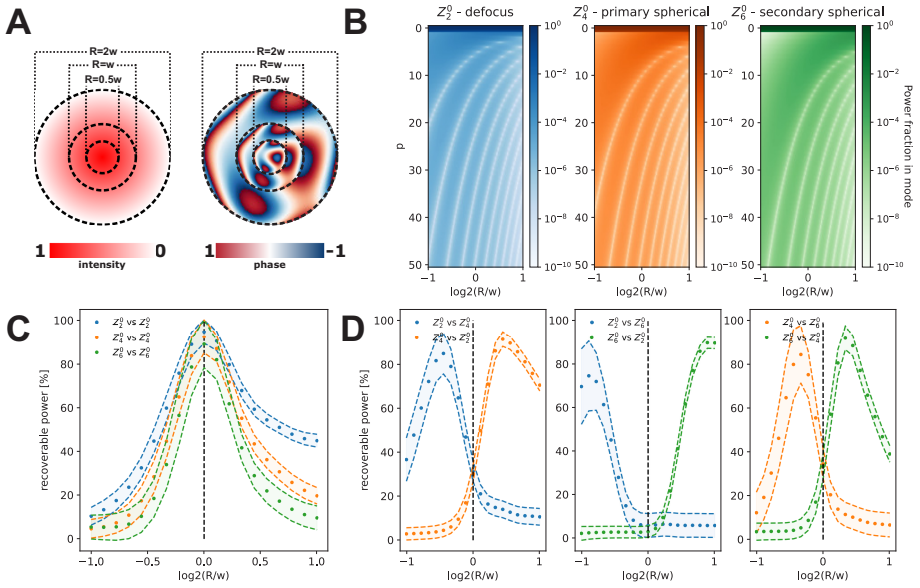


Fig. 1. **A** Cartoon representation of the wavefront dependence on the aberration radius (R). Left: Intensity cross-section. Right: Wavefront amplitude. **B** Dependence of the power coupling into higher order LG-modes on the ratio between the aberration and beam radii (R/w). **C** Dependence of the self-compensation in LG-space on the R/w ratio using the first 25 LG modes. The shaded area represents the uncertainty range (which originates from the finite subset of LG modes considered) between the lower and upper bounds calculated using Eq. (7) and Equation S10 (see Supplement 1 Section S3 for details). The dots represent the expected value. **D** Dependence of cross-compensation between different aberrations on the R/w ratio using the first 25 LG modes.

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3. Direct evaluation of Zernike aberration cross-coupling in Gaussian beams

The framework based on analysing the coupling of aberrated Gaussian beams into higher order LG-beams is very useful, however in this particular application it suffers from the limitations due to the uncertainty of evaluating the crossangle (Fig. 1(D)). To alleviate this problem we propose an alternative formulation based on extending a framework proposed for directly evaluating the crosscoupling between Zernike aberrated Gaussian beams [9,12] to accommodate arbitrary combinations of aberrations (for treatment of strong aberrations see Supplement 1 Section S4):

$$k_{GG}^{Z} = \int_A G^2 \exp\left(ik\sum_{\alpha} Z_\alpha\right) dA \approx 1 + ik\sum_{\alpha} I_\alpha - \frac{k^2}{2}\sum_{\alpha,\beta} I_{\alpha,\beta} \tag{8}$$

where $I_\alpha = \int_A G^2 Z_\alpha dA$ and $I_{\alpha,\beta} = \int_A G^2 Z_\alpha Z_\beta dA$ can be evaluated using equations in Supplementary Sections S5 and S6. As we use the power coupled into the G-mode as a quality metric for beam aberrations we can explicitly calculate the power coupled into the fundamental G-mode:

$$|k_{GG}^{Z}|^2 = 1 - k^2\sum_{\alpha,\beta} I_\alpha I_\beta - \frac{k^2}{2}\sum_{\alpha,\beta} I_{\alpha,\beta} + \frac{k^4}{4}\sum_{\alpha,\beta,\gamma,\delta} I_{\alpha,\beta} I_{\gamma,\delta} \tag{9}$$

One conclusion following from Eq. (9) is that the power coupling dependence on aberration amplitude in the weak aberration regime is described by an n-dimensional 4th order polynomial:

$$p(\mathbf{A}) = \sum_{(i,j,k,l)}^{4}\sum_{\alpha=0}^{4-\alpha}\sum_{\beta=0}^{4-\alpha-\beta}\sum_{\gamma=0}^{4-\alpha-\beta-\gamma}\sum_{\delta=0} A_i^\alpha A_j^\beta A_k^\gamma A_l^\delta c_{\alpha,\beta,\gamma,\delta}^{i,j,k,l} \tag{10}$$

where (i,j,k,l) are sets of distinct indices. For practical applications we differentiate aberrations into passive aberrations (the aberrations present in the system which we cannot control) and active aberrations (the modes our AO element can display in a controlled manner). Using this approach we investigated two simple cases of 1 and 2 active aberrations in a background of passive aberrations. In these cases the polynomials simplify to:

$$P^1(A_i, \mathbf{A_{passive}}) = \sum_{\alpha=0}^{4} A_i^\alpha c_\alpha(\mathbf{A_{passive}}) \tag{11}$$

and

$$P^2(A_i, A_j, \mathbf{A_{passive}}) = \sum_{\alpha=0}^{4}\sum_{\beta=0}^{4-\alpha} A_i^\alpha A_j^\beta c_{\alpha,\beta}(\mathbf{A_{passive}}) \tag{12}$$

where c_α and $c_{\alpha,\beta}$ are the polynomial coefficients (see Supplement 1 Section S7 for explicit form). We used Eq. (11) to show that a low order aberration (Z_4^0) is capable of significantly compensating aberrations even several orders higher (such as Z_{10}^0) when an appropriate R/w ratio is used (Fig. 2(A)). Important to note is the fact that Z_4^0 shows significant cross-compensation for Z_6^0 even at R/w=1 but for higher orders the cross-compensation at R/w=1 is negligible which is in line with our previous results using the LG mode framework [7].

We have also explored more complex systems of aberrations by using Eq. (12) to evaluate how self-compensation influences cross-compensation by modeling the interaction between a $Z_4^0 + Z_6^0$ aberration and a Z_4^0 correction (Fig. 2(B,C)). We can appreciate that the behaviour of the system changes depending on the interaction between the passive aberrations. For a negative amplitude ratio (A_4^0/A_6^0) there is a smooth transition between the states, with an expected reduction in power recovery for aberrations with a higher content of Z_6^0 (as self-compensation is more efficient than

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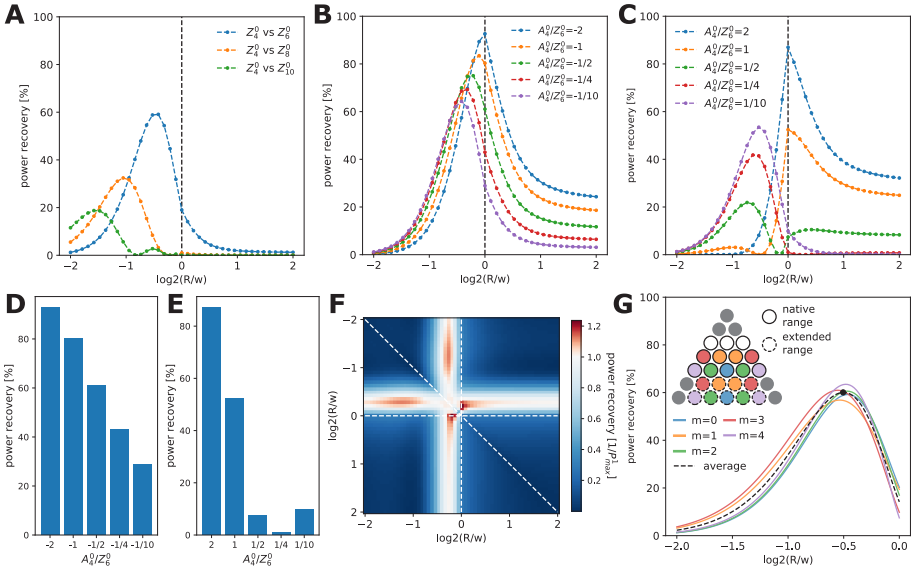


Fig. 2. **A** Power recovery for 3 higher order Z-aberrations (Z_6^0, Z_8^0, Z_{10}^0) when compensated with the lower order Z_4^0 mode (shperical aberration). **B,C** Power recovery for a combined Z_4^0+Z_6^0 aberration with different negative (**B**) and positive (**C**) mixing ratios (A_4^0/A_6^0) when compensated with a Z_4^0 mode. **D,E** Power recovery for a combined Z_4^0+Z_6^0 aberration with different negative (**B**) and positive (**C**) mixing ratios (A_4^0/A_6^0) when compensated with a Z_4^0 mode for R/w=1. **F** Power recovery for a combined Z_6^0+Z_8^0 aberration when compensated with a two Z_4^0 modes of varying R/w. The values on the diagonal are equivalent to using only one Z_4^0 mode (P^1). The scale is normalised to the maximum of the diagonal (P^1_{max}) to show the achievable improvement in power recovery when using two Z_4^0 modes. **G** Pairwise power recovery between subsequent Zernike modes of the same family showing similarities in the usable R/w ratio between different modes.

cross-compensation, Fig. 2(B)). However, for positive amplitude ratios the situation is drastically different (Fig. 2(C)), there is no gradual transition but more of a bi-stable situation where the system seem to switch between 2 optimal R/w ratios. To understand this behaviour better we need to study how the power recovery evolves for R/w=1. We observe that for negative ratios the power recovery slowly decreases, which implies there is no cross-compensation in the background $Z_4^0 + Z_6^0$ aberration (Fig. 2(D)). On the other hand for positive ratios the power recovery has a minimum at around $A_4^0/A_6^0 = 1/4$ (Fig. 2(D)) which indicates that the background aberrations are cross-compensating and nearly balanced for $A_4^0/A_6^0 = 1/4$ in a way where modifying the A_4^0 via aberration correction does not yield a significant improvement. This leads to a bi-stable behaviour where if the Z_4^0 contribution dominates the most efficient strategy is to compensate Z_4^0 by using R/w=1, but if Z_6^0 is dominant it is more efficient to compensate the overall aberration by using $R/w \approx 0.7$.

Finally, by analysing Fig. 2(A) one can begin to hypothesise that using two Z_4^0 aberration at different R/w ratios ($R \approx 0.7$ and $R \approx 0.5$, which in practice would require two separate DMs) might allow efficient simultaneous compensation of a $Z_6^0 + Z_8^0$ aberration further extending the compensation range of DMs. We have tested this hypothesis by again using Eq. (12) and calculated the expected power recovery for different combinations of R/w ratios used (Fig. 2(F)). We note that while the increase in power recovery compared to using 1 DM is only moderate

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(~ 20%) it is striking that it is possible to extend the power recovery of modes 2 orders higher that the one used for correction (Z_4^0 vs Z_8^0).

4. Discussion

Over the past years, the concept of adaptive optics has developed into a powerful method to counteract optical aberrations which allowed for a multitude of imaging related applications most notably in astronomy, but also in biology.

However, hardware limitations of typical AO elements such as deformable mirrors in terms of number of active elements still prohibits robust aberration correction of higher order Zernike aberrations. Our theoretical work explores the possibility of enhancing the capabilities of already existing DMs, therefore allowing efficient correction of higher-order Zernike aberrations using only lower-order modes. We show that it is indeed possible to partially correct aberrations even 3 orders higher than the mode used for correction, as well as to correct several aberrations at the same time.

On a practical note, it is important to point out that the cross-compensation of background aberrations causes complications to the rescaling scheme when used with a single AO element. The proposed framework will work best when two identical AO elements such as deformable mirrors are employed, one operating at R/w=1 to compensate the lower-order aberration within the limits of the AO element and another one working at R/w<1 to compensate for higher order aberrations. Additionally, as determining a proper R/w ratio is crucial for the performance of the proposed method, we would like to point out that in general an optimal R/w ratio can always be found by optimising Eq. (10):

$$\hat{R}_{opt} = argmax(P_{rec}^{\hat{R}}) \tag{13}$$

However, as Equation S21 ascertains orthogonality between aberrations of different families ($m \neq m'$) this approach can be simplified to optimising only pairwise $P_{rec}^{\hat{R}}$ between one active lower order Z-mode and a passive higher order Z-mode:

$$\hat{R}_{opt} = argmax(\sum P_{rec}^{\hat{R}}(Z_n^m, Z_{n+2}^m)) \tag{14}$$

We demonstrate this approach for an exemplary optimisation aimed to extend the range of a DM with a native range of Z3-Z14 (e.g. a DMP40 Thorlabs DM) showing that there is a common R/w value (Fig. 2(G)) that allows to extend the range of the DM by almost full two orders (from 12 to 21 modes) while retaining a 60% power recovery for the higher modes.

Overall, we believe that by analysing the previously unexplored effects of Zernike aberration rescaling this work will enable further studies in the direction of enhancing the capabilities of existing AO devices to tackle increasingly higher order aberrations, which could have significant impact on challenging astronomy and deep-tissue imaging applications.

Funding. Chan Zuckerberg Initiative (2020-225346); European Molecular Biology Laboratory; Deutsche Forschungsgemeinschaft (425902099).

Disclosures. The authors declare that there are no conflicts of interest related to this article.

Data availability. Data is available from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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